CAPSTONE PROJECT

INVESTMENT BANKING POST GRADUATE CERTIFIED COURSE

NOVEMBER 6, 2025

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1. OBJECTIVE

This section documents the complete step-by-step process followed in Microsoft Excel to compute the hedging outcomes, breakeven points, and payoff diagrams for **Mr. A's portfolio** comprising **Ashok Leyland** and **Larsen & Toubro (L&T)**.

The Excel model was designed to ensure analytical accuracy, transparency, and full linkage across all sheets. Each sheet, i.e. **Market Data, Portfolio Details, Hedging Strategies, Breakeven Analysis, and Payoff Diagrams** was interconnected using cell references to maintain a dynamic, real-time recalculation capability whenever any market variable changed.

2. DATA INTEGRATION AND INPUTS

All fundamental market inputs were drawn from the **NSE website** (as of **28th October 2025**) and consolidated into a master sheet titled "**Market Data**."

The data included:

- Current Spot Prices, Futures Prices, Option Strikes, and Premiums (Call and Put)
- Lot Sizes and Expiry Dates for both Ashok Leyland and L&T
- Portfolio Quantities and Purchase Prices for Mr. A's holdings
- A summary input table was created in Excel using cell references for automatic linkage across all other sheets.

This integration ensured that every sheet in the workbook updated automatically whenever new market data was entered.

3. PORTFOLIO SETUP

Mr. A's portfolio comprised long equity positions in two companies:

- Ashok Leyland: 5,000 shares @ ₹180 each
- L&T: 300 shares @ ₹2,600 each

The following calculations were performed in Excel:

Total Exposure = Buy Price × Quantity

- Ashok Leyland: ₹180 × 5,000 = **₹9,00,000**
- L&T: ₹2,600 × 300 = **₹7,80,000**

Additionally, current market values were derived using the Spot Prices from NSE, and Unrealized P/L was calculated as:

Unrealized P/L = (Spot Price − Buy Price) × Quantity

This established the portfolio's existing mark-to-market position, providing a foundation for subsequent hedging analysis.

4. SPOT PRICE GENERATION

To analyse hedge performance under different market movements, a range of future spot prices was generated for each stock.

The range covered approximately **± 20%** from the current market price, with **2% increments** between each level.

Excel formula used:

Spot Price (i) = Current Spot×(1+0.02×i)

Each row in the scenario table represented a possible market outcome, allowing sensitivity analysis of both stock and derivative payoffs.

Separate columns were created for Ashok Leyland and L&T, maintaining consistent scaling across the sheets.

5. STOCK PROFIT/LOSS COMPUTATION

The unhedged position for each stock was computed using the formula:

Stock P/L = (Spot Price at Expiry−Buy Price) × Quantity

This calculation was carried out for every hypothetical spot price in the scenario range.

Example (Ashok Leyland):

At ₹140, P/L = (140 − 180) × 5,000 = **–₹2,00,000**

Example (L&T):

At ₹4,000, P/L = (4,000 − 2,600) × 300 = **₹4,20,000**

These results represent the base case (**unhedged exposure**), which was used as a reference for evaluating futures and options hedging performance.

6. FUTURES HEDGE CALCULATION

For each stock, a short futures position was assumed to hedge the long equity exposure.

The formula applied:

Futures P/L = (Futures Sell Price−Spot Price at Expiry) × Lot Size

The combined payoff was then computed as:

Net P/L (Stock + Futures) = Stock P/L + Futures P/L

Illustration:

- **Ashok Leyland Futures @ ₹140.51**

If spot falls to ₹130 → Futures gain = (140.51 − 130) × 5,000 = ₹52,550

Stock loss = –₹2,50,000

Net Result = +₹1,150 (constant across prices)

- **L&T Futures @ ₹3,928**

At **any expiry level**, the **Net P/L = –₹2,220**, reflecting a minor basis difference.

The Excel table validated that the Futures Hedge flattened the P/L curve, confirming a perfect hedge where directional risk was nearly eliminated.

7. OPTIONS HEDGE CALCULATION

A **protective put hedge** was implemented for both positions using current NSE premiums and strikes.

The payoff was modelled using the formula:

Option P/L = max (0, Strike Price−Spot Price) × Lot Size − (Premium Paid×Lot Size)

The combined payoff after hedging was computed as:

Net P/L (Stock + Option) = Stock P/L + Option P/L

- **Example – Ashok Leyland (Put Strike ₹150, Premium ₹9.30):**

If Spot = ₹130 → Option Payoff = (150 − 130) × 5,000 = ₹1,00,000

Net Option P/L = ₹1,00,000 − ₹46,500 = ₹53,500

Stock P/L = –₹2,50,000

Net Combined = –₹1,96,500 (loss capped)

- **Example – L&T (Put Strike ₹4,000, Premium ₹75.30):**

If Spot = ₹3,500 → Option Payoff = (4,000 − 3,500) × 300 = ₹1,50,000

Net Option P/L = ₹1,50,000 − ₹22,590 = ₹1,27,410

Stock P/L = (3,500 − 2,600) × 300 = ₹2,70,000

Net Combined = ₹3,97,410 (protected profit)

This clearly demonstrated that the put hedge set a guaranteed floor value, allowing downside protection with upside potential intact.

8. BREAKEVEN POINT CALCULATION

Two breakeven concepts were computed for each stock:

- **Option-only Breakeven**

Breakeven Price = Strike Price − Premium Paid

Ashok Leyland: 150 − 9.30 = **₹140.70**

L&T: 4,000 − 75.30 = **₹3,924.70**

This indicates the price where the option itself starts yielding a net profit for the buyer.

- **Protective Put Breakeven (Stock + Put)**

Breakeven Price = Buy Price + Premium Paid

Ashok Leyland: 180 + 9.30 = **₹189.30**

L&T: 2,600 + 75.30 = **₹2,675.30**

This level marks the total price at which the combined position (equity + put) begins to yield positive returns, considering the option cost.

9. PAYOFF DIAGRAM CONSTRUCTION

To visually demonstrate hedge efficiency, dynamic payoff charts were constructed for both instruments and both stocks.

- **Futures Hedge Payoff Diagram:**

Data Source: Spot Prices and Net (Stock + Futures) P/L

Chart Type: Line with Markers

X-axis: Spot Price (₹)

Y-axis: Net Profit/Loss (₹)

Output: A near-flat line indicating a neutralized price risk.

- **Options Hedge Payoff Diagram:**

Data Source: Spot Prices, Stock P/L, and Net (Stock + Option) P/L

Chart Type: Line with Two Series

X-axis: Spot Price (₹)

Y-axis: Net Profit/Loss (₹)

- **Visual Markers:**

Vertical dashed line at the Strike Price

Annotation for Breakeven Point and Max Loss (Premium Paid)

The payoff diagram displayed the classic “hockey stick” shape; flat on the downside, rising on the upside; showing the effectiveness of protective puts in limiting losses.

10. FINAL OUTCOME

The Excel-based model effectively demonstrated:

- How Futures Hedges stabilize returns through symmetric exposure offsetting.
- How Options Hedges cap downside risk while preserving upside potential.
- Accurate identification of breakeven points using real NSE premiums and expiry data.
- Real-time interconnectivity between Market Data, Portfolio, and Hedge Analysis sheets.

This dynamic spreadsheet model not only quantifies financial exposure but also visually communicates the practical application of hedging tools in professional portfolio risk management.